

Machine Learning II

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Business context

Employee Promotion means the ascension of an employee to higher ranks, this aspect of the job is what drives employees the most. And consists of details of all the employees in the company working last year and if It got promoted or not, but every time this process gets delayed due to so many details available for each employee - it gets difficult to compare and decide.

Objective

For the upcoming appraisal cycle, the HR team wants to utilize the stored data and leverage machine learning to make a model that will predict if a person is eligible for promotion or not. , as a data scientist at JMD company, need to come up with the best possible model that will help the HR team to predict if a person is eligible for promotion or not.

Data Description

Variable Name	Data Type	Category	Description
employee_id	Numerical / ID	Identifiers	Unique ID for the employee
department	Categorical	Organizational	Department of employee
region	Categorical	Demographics	Region of employment (unordered)
education	Categorical	Demographics	Education Level
gender	Categorical	Demographics	Gender of Employee
recruitment_channel	Categorical	Organizational	Channel of recruitment for employee
no_of_trainings	Numerical	Performance	Number of other trainings completed in the previous year

age	Numerical	Demographics	Age of Employee
previous_year_rating	Numerical	Performance	Employee Rating for the previous year
length_of_service	Numerical	Organizational	Length of service in years
awards_won	Binary	Performance	If awards won during previous year
avg_training_score	Numerical	Performance	Average score in current training evaluations
is_promoted	Binary	Target	Recommended for promotion

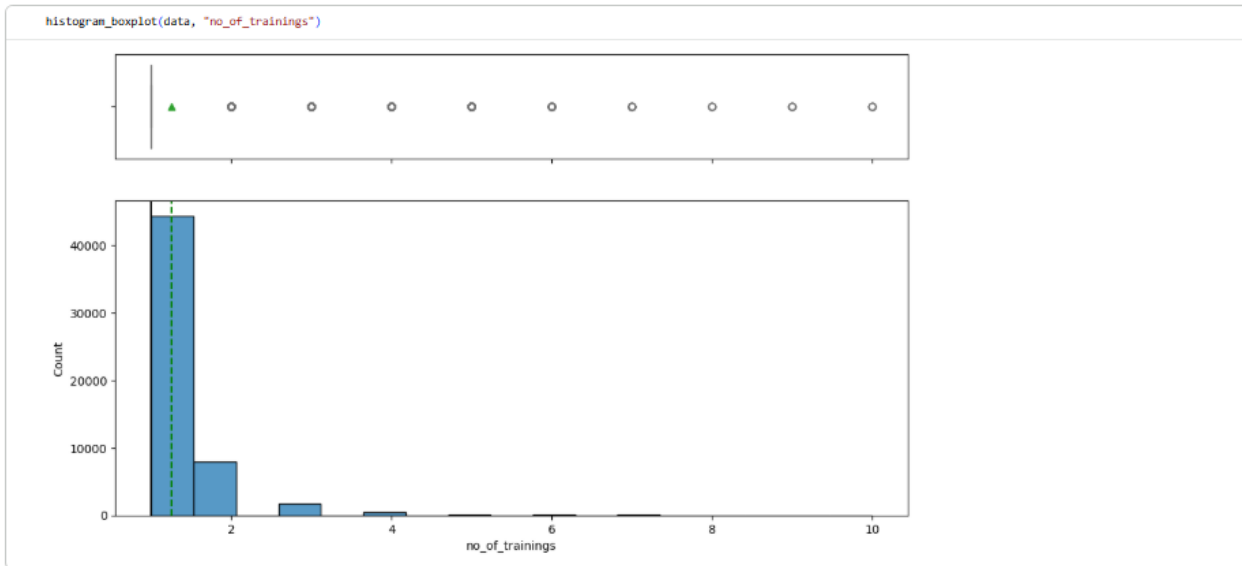
Exploratory Data Analysis

Univariate analysis

Image :1

Observations on No. of Trainings

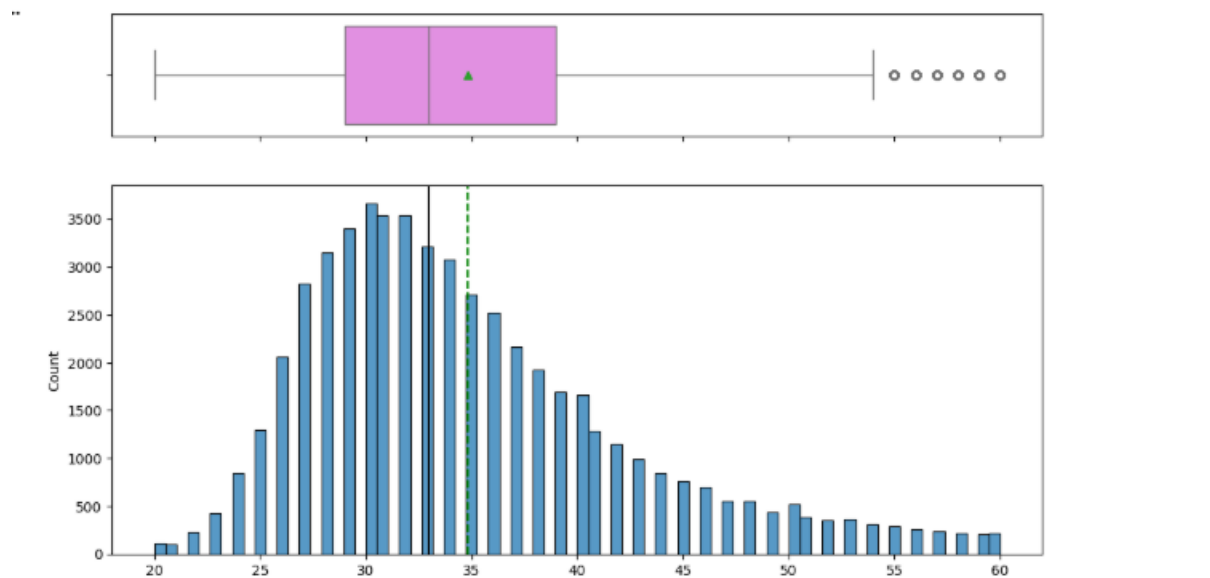
Observations on No. of Trainings



- **Heavy Right Skew:** The vast majority of employees take exactly 1 training per year. The count drops off drastically for 2, and becomes an extreme rarity beyond 3.
- **The "Baseline" Majority:** Over 40,000 employees sit in the minimum bin (1). This is the standard operational baseline for the company.
- **Statistical Outliers:** The boxplot identifies anyone taking 2 or more trainings as a statistical outlier (represented by the open circles extending all the way to 10).
- **The Mean vs. Median Shift:** The median (black line) is pinned at 1 . The mean (green triangle/dashed line) is pulled slightly to the right (1.2) due to the small group of hyper-proactive employees taking 4+ courses.

Image :2

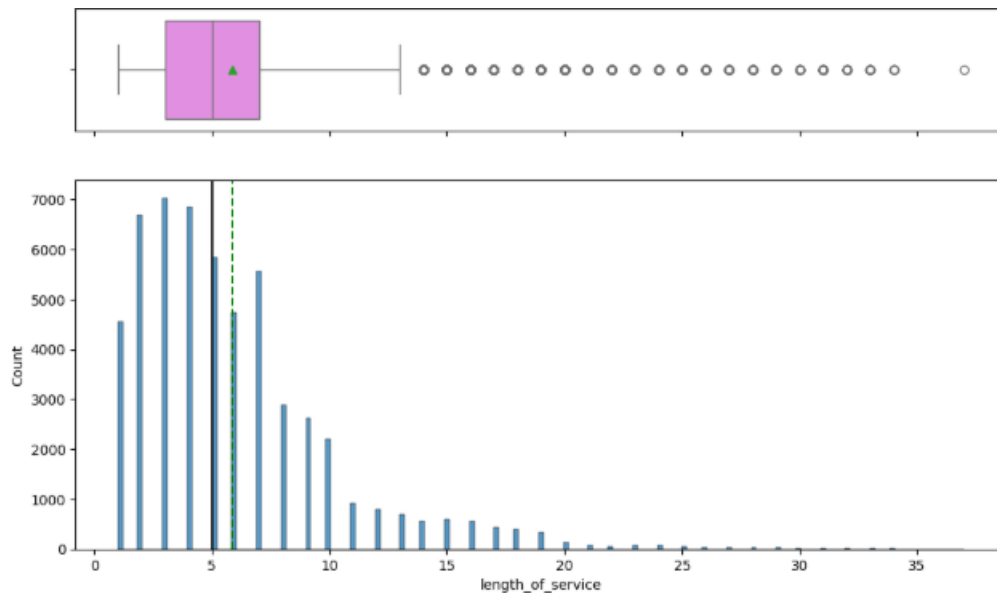
Observations on Age



- **Right-Skewed Workforce:** The employee distribution peaks early around ages 30 to 33 (the median is marked by the solid black line at 33). The population thins out steadily as age increases toward 60.
- **Young & Mid-Level Concentration:** The majority of the workforce (represented by the bulk of the pink box) sits between 29 and 39 years old (the Interquartile Range, or IQR). This indicates a highly active, mid-career demographic.
- **Senior Outliers:** The boxplot flags employees aged 55 and above as statistical outliers (open circles). These represent long-tenured employees or senior leadership, making up a much smaller fraction of the overall dataset.
- **Mean vs. Median Gap:** The mean age (green triangle/dashed line at 35) is pulled higher than the median (33) due to the long right tail of older employees.

Image :3

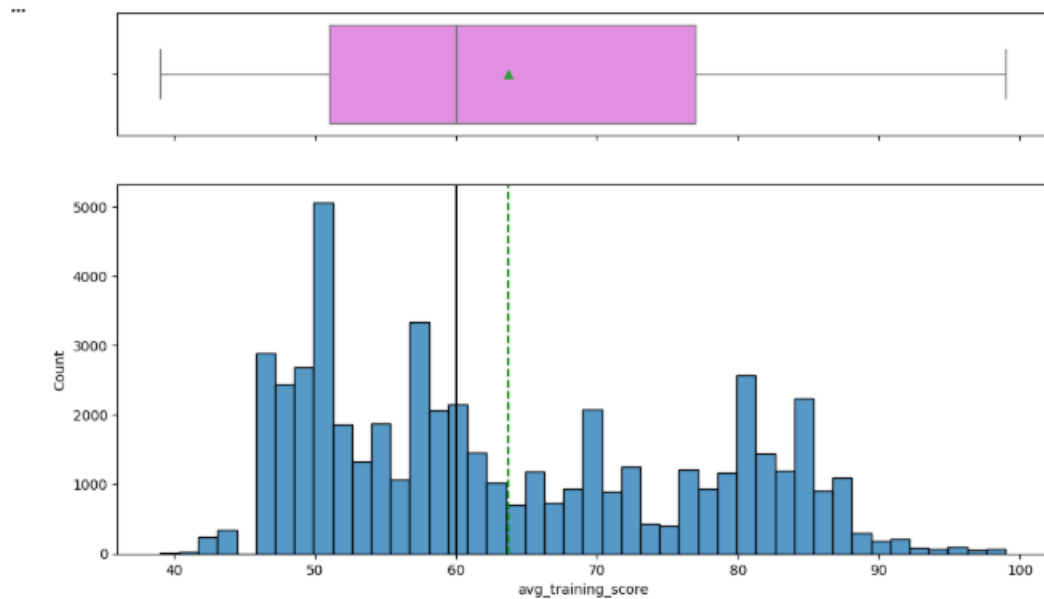
Observations on Length of Service



- **High Early-Career Turnover/Concentration:** The distribution peaks sharply within the 2 to 5 years range (with the median marked by the black line at exactly 5 years). This indicates a workforce heavily populated by early-to-mid career employees.
- **The "Sweet Spot" Box:** Half of the entire workforce (the pink Interquartile Range box) falls between 3 and 7 years of service. This is the core operational demographic of the company.
- **Heavy Right Skew (Tenure Outliers):** The distribution trails off severely after 10 years. The boxplot flags anyone with more than 13 years of service as a statistical outlier (the long string of open circles extending all the way to 37 years).
- **The Attrition Curve:** The rapid, exponential drop-off in counts between 7 and 15 years suggests natural corporate attrition or that the company has scaled up hiring significantly only in recent years.

Image :4

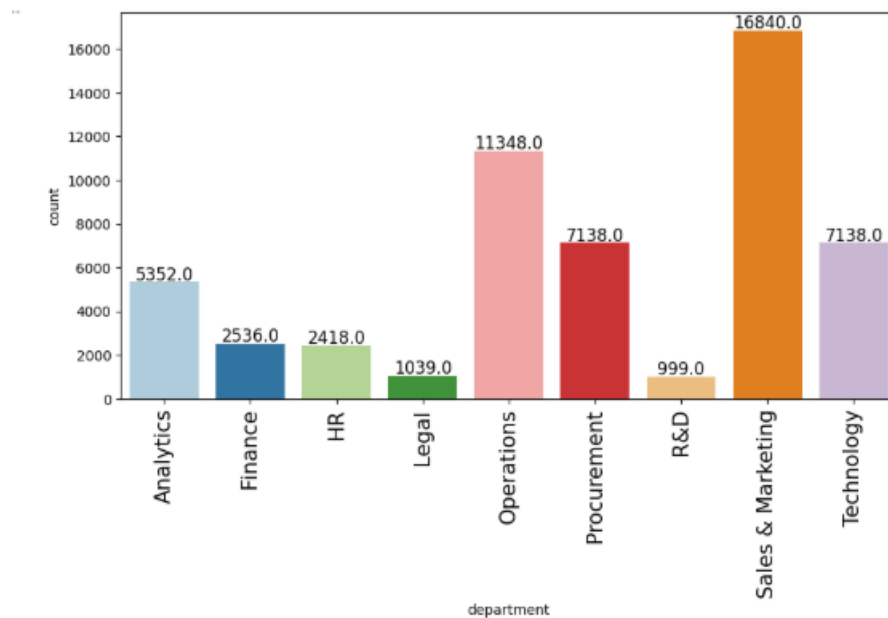
Observations on Average Training Score



- **Multimodal Distribution (Multiple Peaks):** Unlike the previous variables, this chart shows a wavy, multi-peaked distribution with distinct clusters around 50, 58, 70, and 80–85.
- **The Departmental Fingerprint:** This multi-peak pattern is a classic indicator that multiple distinct groups are mixed together. It confirms our earlier hypothesis: different departments have completely different evaluation metrics or exam difficulties (e.g., Sales testing peaks at 50, while IT/Tech testing peaks at 80).
- **Perfect Spread, No Outliers:** The pink boxplot shows a wide Interquartile Range (IQR) stretching from roughly 51 to 76, with the median pinned at 60 and the mean at 64. Notably, there are zero outlier dots—the entire range from 40 to 100 is filled with valid, continuous employee scores.

Image : 5

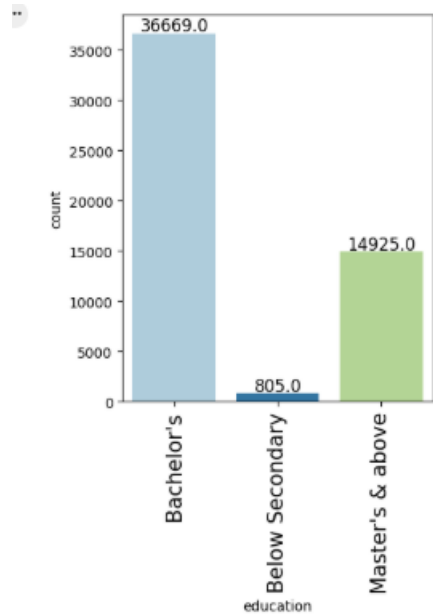
Observations on Department



- **Heavy Revenue/Customer-Facing Focus:** The organization is overwhelmingly dominated by Sales & Marketing (16,840) and Operations (11,348). Together, these two departments comprise more than half of the entire workforce.
- **Mid-Tier Technical Support:** Procurement (7,138), Technology (7,138), and Analytics (5,352) form a solid secondary layer of operational and technical infrastructure.
- **Niche Corporate G&A:** The specialized support divisions—Finance, HR, Legal, and R&D—have the smallest headcount footprint, with R&D being the smallest at just 999 employees.

Image : 6

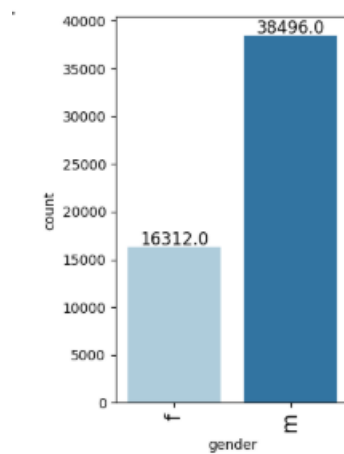
Observations on Education



- **Highly Educated Workforce:** The vast majority of the company's workforce holds an undergraduate degree, with Bachelor's (36,669) making up the overwhelming baseline class.
- **Significant Post-Graduate Presence:** A very healthy secondary tier of employees holds advanced degrees, with Master's & above (14,925) accounting for roughly 28% of the visible distribution.

Image : 7

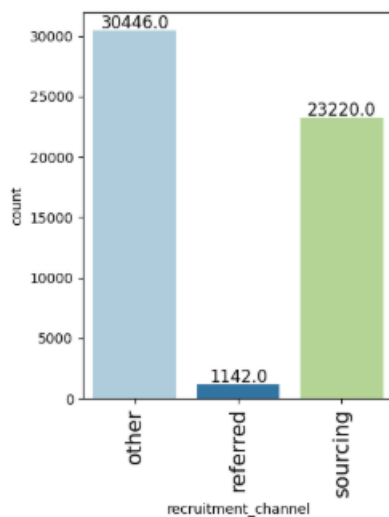
Observations on Gender



- **Pronounced Imbalance:** The workforce shows a significant gender skew, with Male employees (38,496) outnumbering Female employees (16,312) by a ratio of roughly 2.4:1 . Males make up approximately 70\% of the total dataset.
- **No Missing Data:** Adding these two categories (16,312 + 38,496 = 54,808) matches the total row count of r dataset perfectly, confirming there are no missing or unrecorded gender value

Image : 8

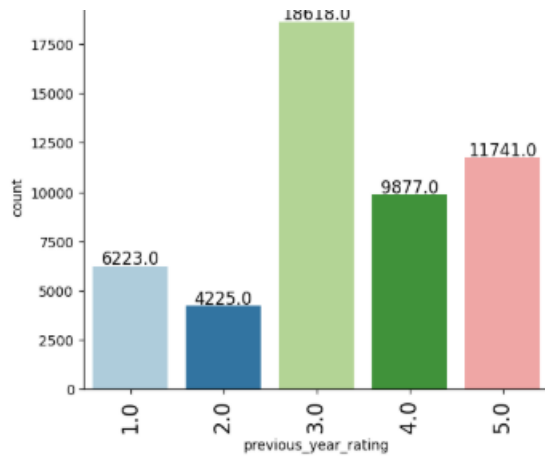
Observations on Recruitment Channel



- **Dominant Channels:** The vast majority of candidates come through "other" (30,446.0) and "sourcing" (23,220.0) channels. Together, these two categories make up over 95% of the total recruitment volume shown.
- **Low Referral Volume:** Employee or professional referrals ("referred") contribute a tiny fraction of the total count, coming in at just 1,142.0

Image : 9

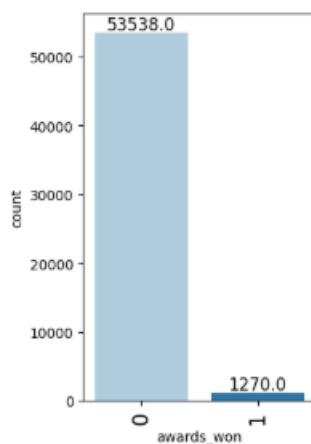
Observations on Previous Year Rating



- **The Average Peak:** The distribution peaks sharply at a rating of 3.0 (18,618.0 employees). This represents the largest single group, which is typical for standard corporate performance distributions where the majority meet expectations.
- **Positive Lean:** There is a strong leaning toward higher performance overall. High performers—ratings 4.0 (9,877.0) and 5.0 (11,741.0)—account for over 21,600 employees. Interestingly, there are more top-tier (5.0) performers than mid-high (4.0) performers.
- **Low Performers:** Ratings 1.0 (6,223.0) and 2.0 (4,225.0) represent the smallest segments of the workforce, though a count of over 6,000 at a 1.0 rating is still a significant cohort requiring attention.

Image : 10

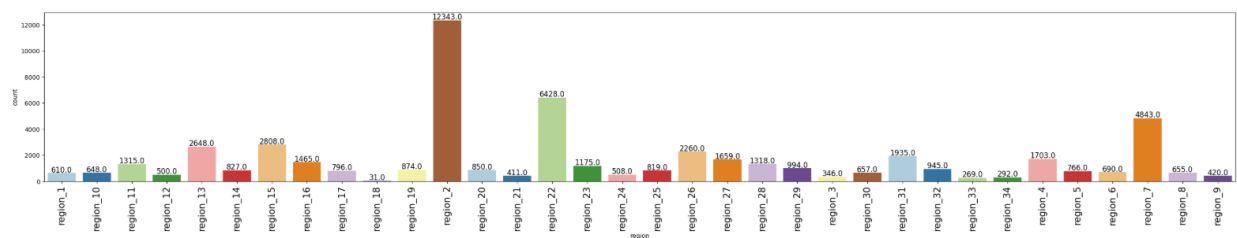
Observations on Awards Won



- Severe Class Imbalance: The dataset is heavily skewed toward employees who have not won an award. A staggering 53,538.0 employees have a value of 0 (no awards won).
- Elite Minority: Only 1,270.0 employees have won an award (value of 1). This accounts for roughly 2.3\% of the total workforce represented in this chart.

Image : 11

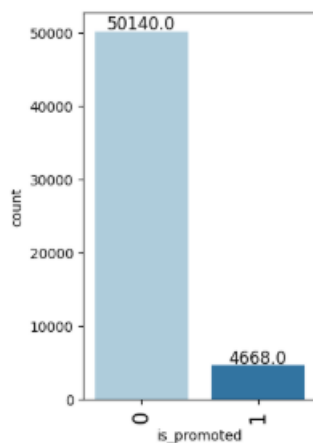
Observations on Region



- The Dominant Peak: region_2 is the absolute outlier with the highest count of 12,343.0, followed by region_22 (6,428.0) and region_7 (4,843.0).
- The Lowest Point: region_18 has the least activity with a count of just 31.0.
- Data Imbalance: The chart shows a heavily skewed distribution; a small handful of regions drive the vast majority of the total volume.

Image : 12

Observations on target variable



- Severe Class Imbalance: The dataset is heavily skewed toward employees who were not promoted (0), totaling 50,140.
- Low Promotion Rate: Only 4,668 employees were promoted (1).
- The Takeaway: Promoted individuals make up roughly 8.5% of the total dataset.

Bivariate Analysis

Image : 13

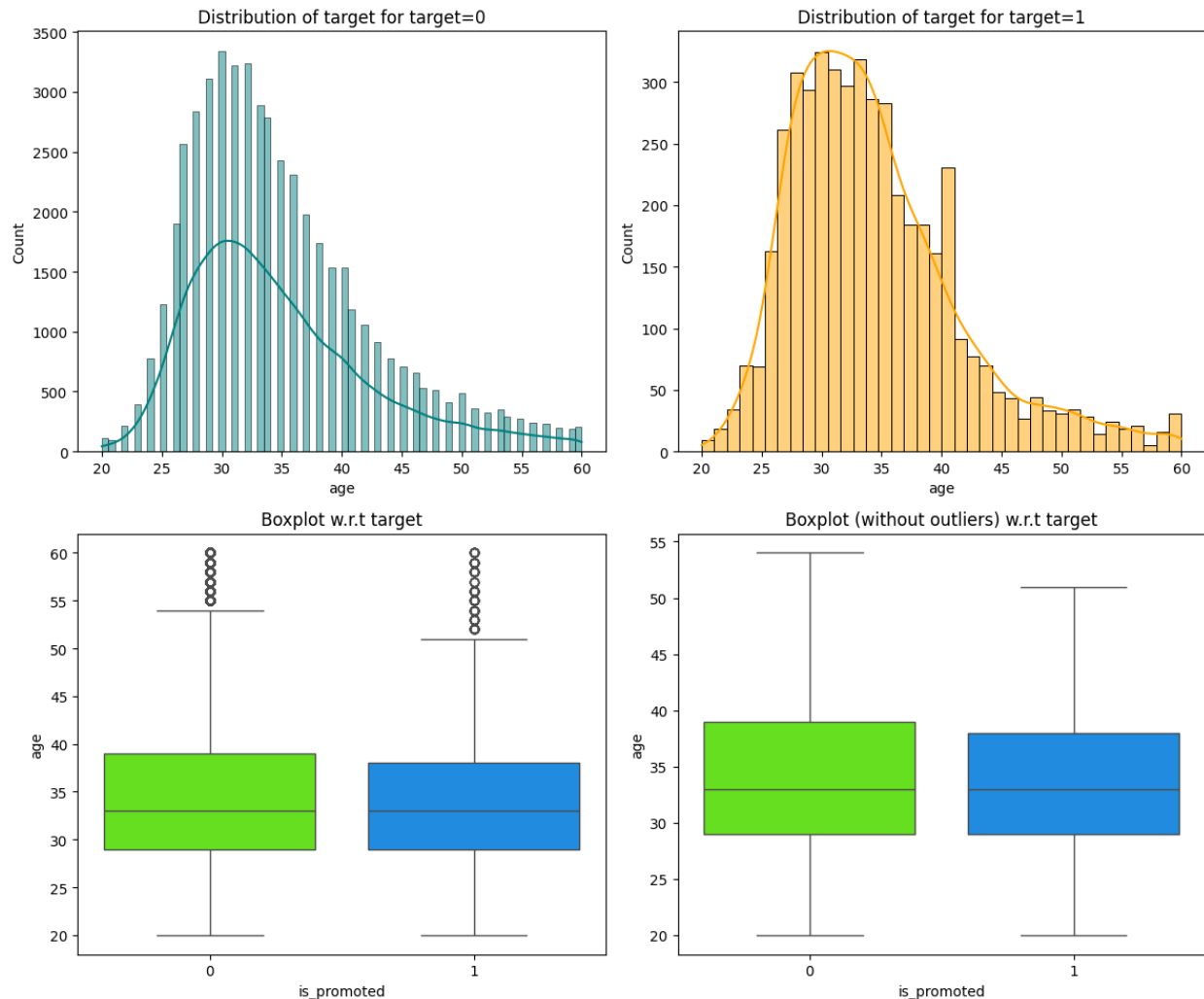


- Top Driver (avg_training_score): Scores between 80–100 show a massive, distinct cluster of promotions. This is the strongest predictor.

- Quick Boost (awards_won): Employees who won an award (1) have a drastically higher promotion rate than those who didn't.
- Redundancy (age vs. length_of_service): These two features are highly correlated linearly. Keep an eye on multicollinearity if are using a linear model.
- Low Impact: no_of_trainings and previous_year_rating show heavy overlap between classes, meaning It have low standalone predictive power.

Image : 14

Target variable vs Age

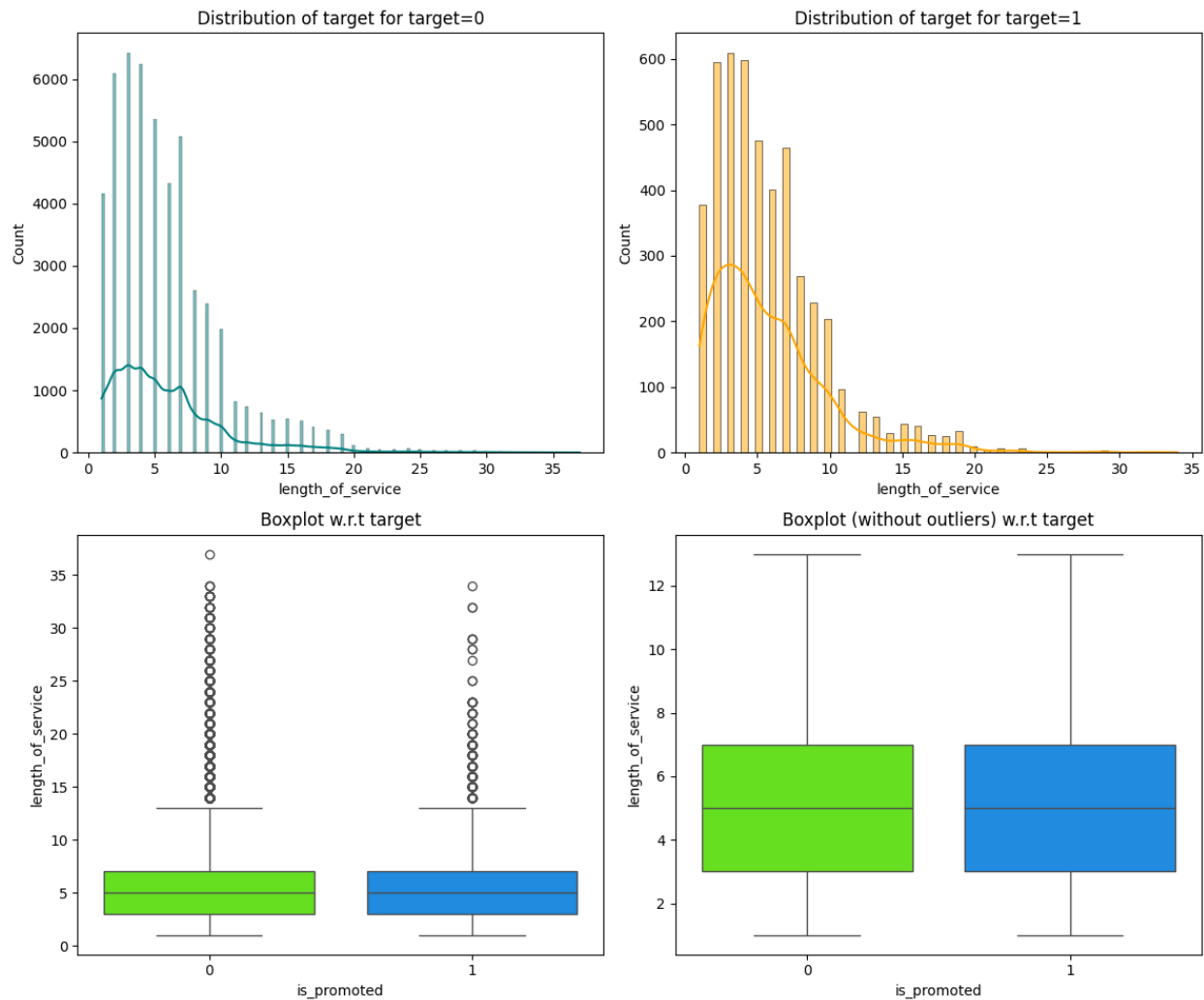


- Nearly Identical Age Distributions: Both the histograms and boxplots reveal that the age distributions for promoted (1) and non-promoted (0) employees are virtually identical. Both groups peak around 30 to 33 years old and are right-skewed.
- No Age Discrimination/Bias: The median age (the center line in the boxplots) sits exactly at 33 years old for both categories. This indicates that age is not a major determining factor or constraint for promotion eligibility.

- Senior Outliers: The boxplot on the bottom-left shows several outliers above the age of 52–54 for both groups. When removing these outliers (bottom-right plot), the upper range for the non-promoted group sits slightly higher (~ 54) compared to the promoted group (~ 51).

Image : 15

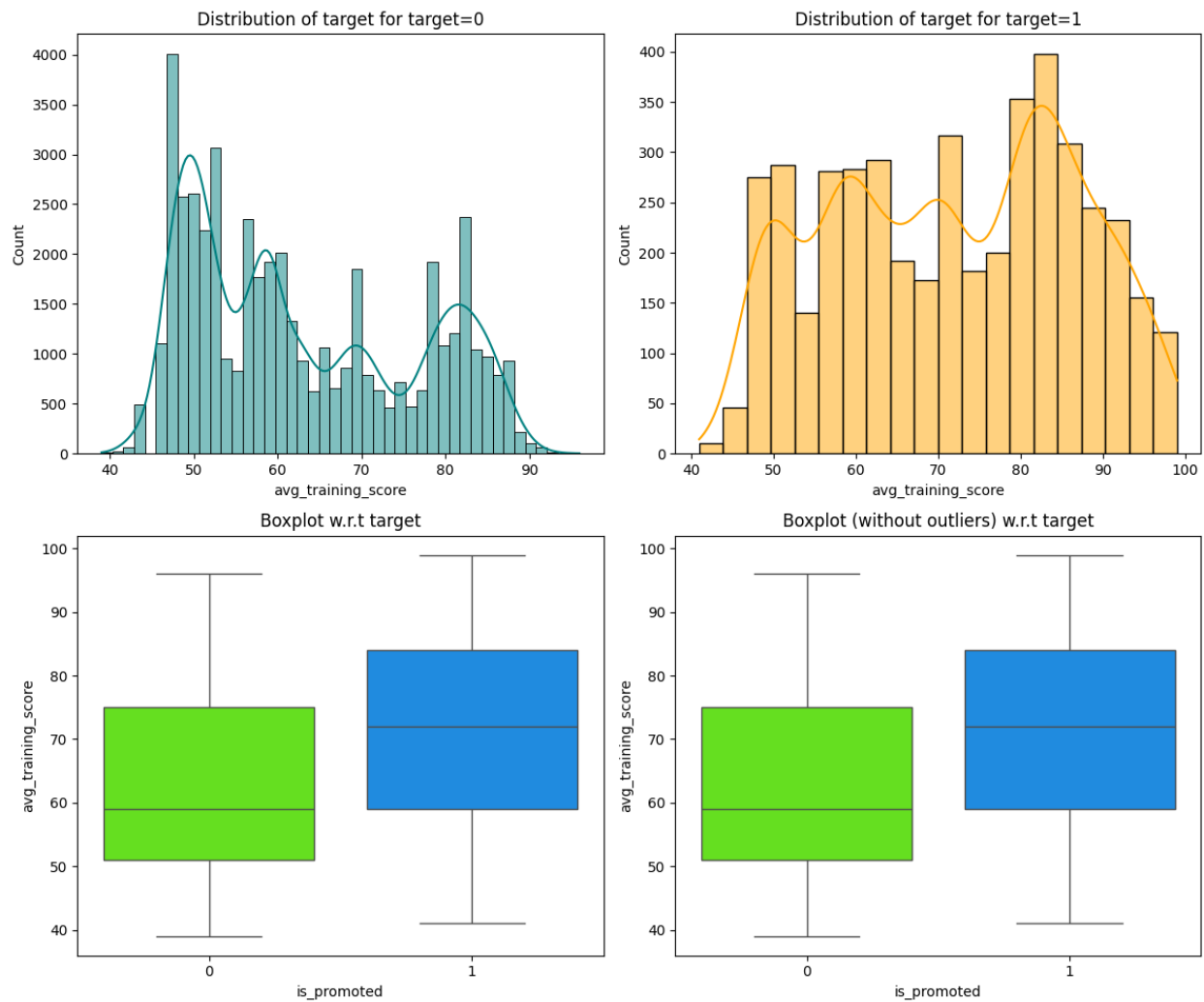
Target variable vs Length of Service



- Identical Tenures: The length_of_service distributions for both promoted (1) and non-promoted (0) employees are practically identical. Both peak intensely between 2 to 5 years of service and are heavily right-skewed.
- Overlapping Medians: The median length of service is exactly 5 years for both groups (shown by the center lines in the boxplots). This means senior tenure is not a strict requirement for moving up.
- Heavy Outliers: The bottom-left boxplot highlights massive outlier structures extending up to 34–37 years for both categories. When these outliers are removed (bottom-right), the core interquartile ranges (IQR) perfectly overlap between 3 and 7 years.

Image : 16

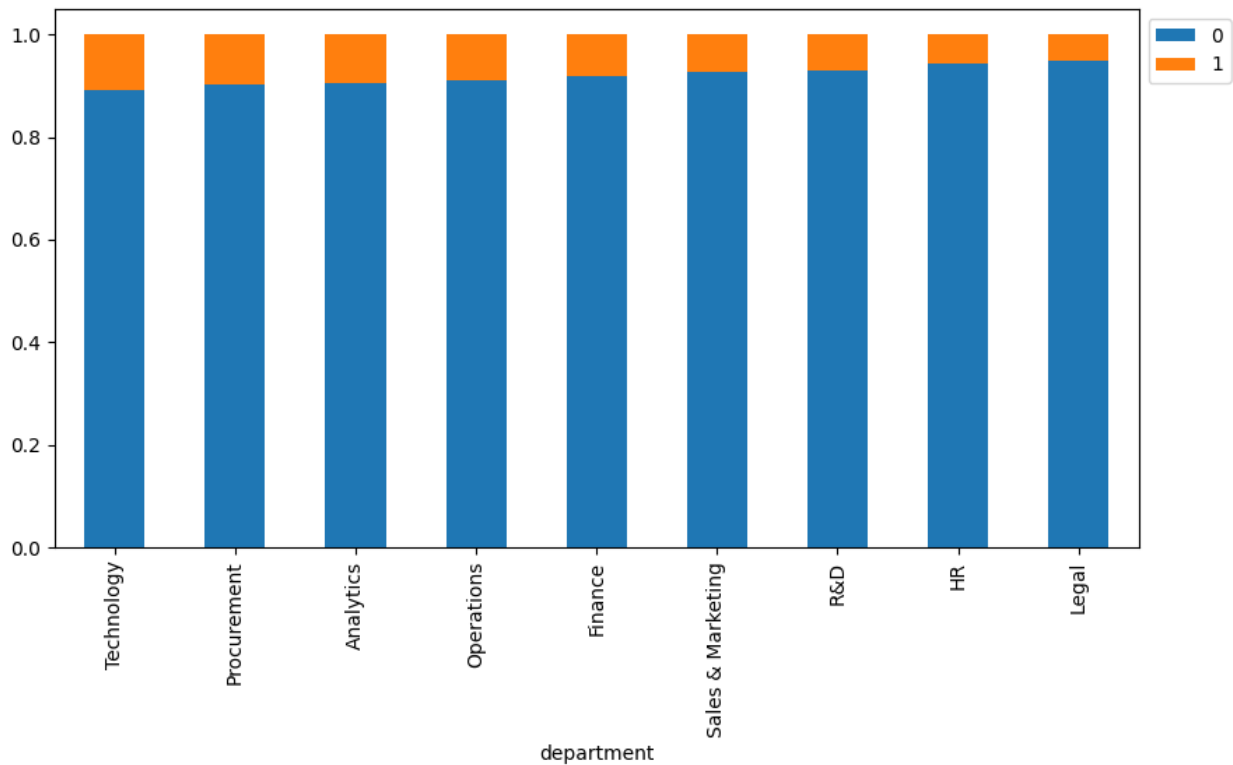
Target variable vs Average Training Score



- **Strong Feature Separation:** Unlike age or tenure, avg_training_score shows a noticeable shift between the two groups. Promoted employees (1) consistently score higher.
- **Higher Median Performance:** The median score for promoted employees sits significantly higher at **72**, compared to a median score of **59** for non-promoted employees.
- **The High-Score Shift (Top Right Histogram):** For the promoted group, the distribution peaks aggressively between **80 and 85**. While non-promoted employees peak sharply around **48-50** and taper off, the promoted group maintains sustained high volume across higher score brackets up to nearly 100.
- **Zero Outliers:** The left and right boxplots are completely identical, confirming there are no statistical outliers in either distribution.

Target variable vs Department

is_promoted	0	1	All
department			
All	50140	4668	54808
Sales & Marketing	15627	1213	16840
Operations	10325	1023	11348
Technology	6370	768	7138
Procurement	6450	688	7138
Analytics	4840	512	5352
Finance	2330	206	2536
HR	2282	136	2418
R&D	930	69	999
Legal	986	53	1039



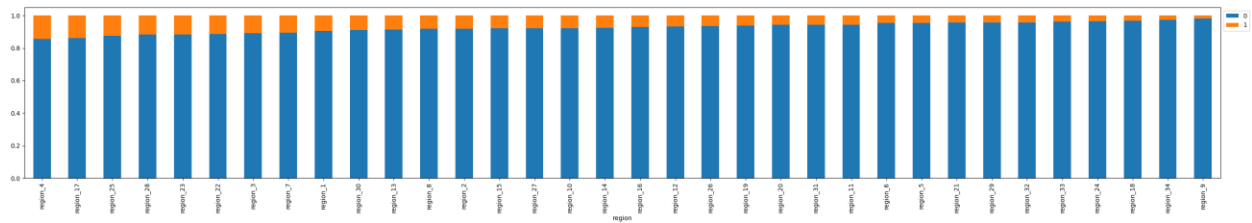
- **Uniform Promotion Rates:** Despite massive differences in headcount across departments, the actual proportion of promotions (the orange bars) remains highly consistent, hovering right around 5% to 11% across the board.
- **Top Promotion Rates:** Technology leads with the highest relative promotion rate at roughly 10.7% (768 / 7138), followed closely by Procurement at 9.6% (688 / 7138) and Analytics at 9.5% (512 / 5352).
- **Lowest Promotion Rates:** Legal and HR have the lowest relative promotion rates, coming in at approximately 5.1% (53 / 1039) and 5.6% (136 / 2418) respectively.

- Volume vs. Percentage: Sales & Marketing has by far the highest absolute number of promotions (1,213), but because it is an enormous department, its actual promotion rate sits in the middle of the pack at 7.2%.

Image : 18

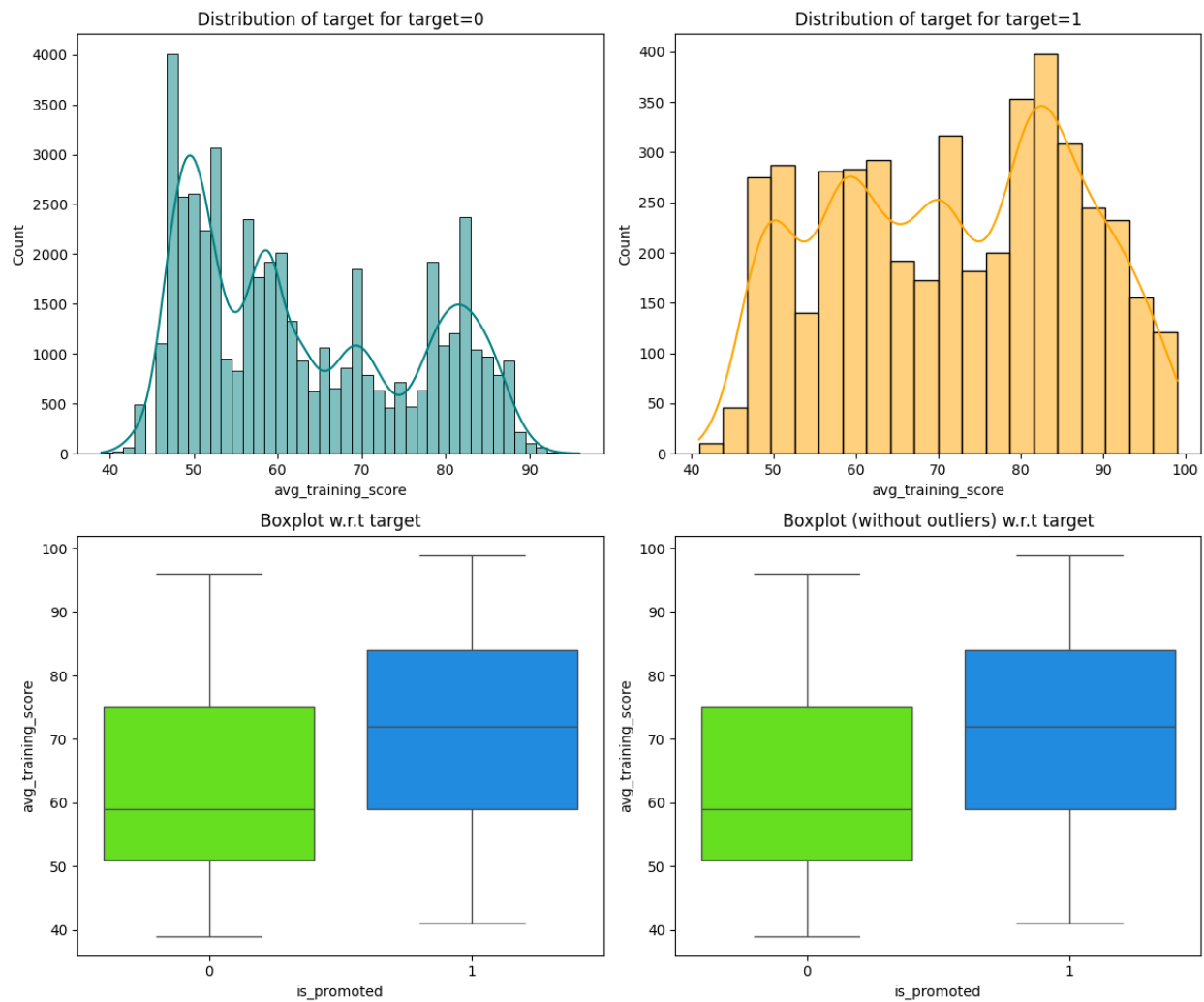
Target variable vs Region

is_promoted	0	1	All
region			
All	50140	4668	54808
region_2	11354	989	12343
region_22	5694	734	6428
region_7	4327	516	4843
region_4	1457	246	1703
region_13	2418	230	2648
region_15	2586	222	2808
region_28	1164	154	1318
region_26	2117	143	2260
region_23	1038	137	1175
region_27	1528	131	1659
region_31	1825	110	1935
region_17	687	109	796
region_25	716	103	819
region_16	1363	102	1465
region_11	1241	74	1315
region_14	765	62	827
region_30	598	59	657
region_1	552	58	610
region_8	602	53	655
region_19	821	53	874
region_10	597	51	648
region_20	801	49	850
region_29	951	43	994
region_32	905	40	945
region_3	309	37	346
region_5	731	35	766
region_12	467	33	500
region_6	658	32	690
region_21	393	18	411
region_24	490	18	508
region_33	259	10	269
region_34	284	8	292
region_9	412	8	420
region_18	30	1	31



- **Highly Consistent Promotion Rates:** Just like the departmental analysis, the *percentage* of promotions (the orange bars) remains uniform across virtually all regions, hovering steadily between 7% and 14%. Regional geography does not strictly favor or penalize an employee's career progression.
- **Top Performing Region by Volume:** region_2 yields the highest absolute number of promotions by a landslide with 989 individuals promoted. However, because its overall workforce size is massive (12,343), its promotion rate is a standard 8.0%.
- **Highest Relative Promotion Rate:** region_4 has the highest conversion efficiency. Out of 1,703 employees, 246 were promoted, yielding a top-tier regional promotion rate of roughly 14.4%.
- **Lowest Relative Promotion Rate:** region_9 sits at the bottom of the spectrum with a promotion rate of just 1.9% (8 / 420).

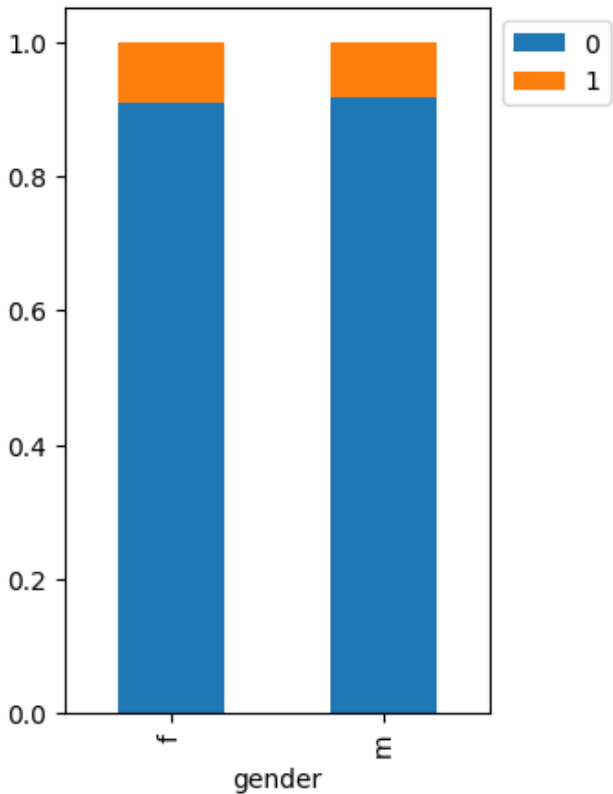
Target variable vs Education



- **Strong Feature Separation:** `avg_training_score` shows a distinct upward shift for promoted employees (1).
- **Higher Distribution Peaks:** Non-promoted employees (0) peak sharply around a score of 48–50 and drop off. Conversely, promoted employees sustain high counts across higher scores, aggressively peaking between 80 and 85.
- **Significantly Higher Median:** The median training score for promoted employees sits at 72, compared to a much lower median of 59 for those not promoted.
- **No Statistical Outliers:** The boxplots with and without outliers are completely identical, confirming that the data distribution has clean boundaries on both ends.

Target variable vs Gender

is_promoted	0	1	All
gender			
All	50140	4668	54808
m	35295	3201	38496
f	14845	1467	16312



- Gender Imbalance: The workforce is heavily male-dominated (70% male vs. 30% female).
- Equal Opportunity: The promotion rate is nearly identical for both groups— 9.0% for women and 8.3% for men.

Correlation Heatmap



- Strongest Relationship: age and length_of_service have a strong positive correlation (0.66)—older employees naturally tend to have longer tenures.
- Top Promotion Drivers: Winning an award (awards_won: 0.20), a high training score (avg_training_score: 0.18), and a strong performance rating (previous_year_rating: 0.16) are the strongest positive correlates with being promoted (is_promoted).
- Negligible Factors: Demographics and tenure (age, length_of_service, and no_of_trainings) show virtually zero correlation with promotion, meaning seniority alone does not guarantee a step up.

Data Preprocessing

A dataset shape of (54808, 13) means we have 54,808 employee records (rows) and 13 distinct attributes (columns), including the target variable is_promoted.

- previous_year_rating: 4,124 missing values
- education: 2,409 missing values
- avg_training_score: 2,560 missing values
- All other columns: 0 missing values
- Categorical Imputation: education is correctly imputed using the most_frequent strategy (mode), which matches its non-numerical nature.

- Numerical Imputation: `previous_year_rating` and `avg_training_score` are imputed using the median strategy. Relying on the median rather than the mean protects the imputation from being heavily skewed by potential extreme outliers.
- `employee_id` (54,808): Identical to row count. . Drop it.
- `department` (9), `recruitment_channel` (3), `gender` (2): Low variety. One-Hot Encode it cleanly.
- `education` (3): Strict hierarchy. Ordinal Encode (1, 2, 3) to maintain rank.
- `region` (34): High variety. Use Target Encoding to prevent a massive column bloat.
- `age` (41), `length_of_service` (35): Wide continuous spans. Scale if using linear models.
- `is_promoted` (2): Binary target. Expect heavy class imbalance. Track F1-Score, not accuracy.

Building the model

Model Building

Before Oversampling, counts of label 'Yes': 3734

Before Oversampling, counts of label 'No': 40112

After Oversampling, counts of label 'Yes': 40112

After Oversampling, counts of label 'No': 40112

After Oversampling, the shape of `train_X`: (80224, 52)

After Oversampling, the shape of `train_y`: (80224,)

- Before: The minority class (Yes) made up only 8.5% of the original training data (3,734 out of 43,846 rows). Left unaddressed, the models would have developed a heavy bias toward predicting No.
- After: Both classes are now perfectly matched at 40,112 rows each (a 50:50 ratio). The models will now weight errors on both classes equally during training.
- The total size of the training partition has nearly doubled, expanding from 43,846 records to 80,224 records.
- Expect training times to increase slightly for ensemble models (like XGBoost and Gradient Boosting) due to this influx of data.

- The shapes line up perfectly: train_X has 80,224 rows across 52 features, matching the 80,224 labels in train_y.

Cross-Validation Cost:

Bagging: 0.7038584587045673

Random Forest: 0.6783220856848068

AdaBoost: 0.5964855874794006

Gradient Boosting: 0.6958158413829658

XGBoost: 0.7229737546792171

Validation Performance:

Bagging: 0.2948717948717949

Random Forest: 0.2606837606837607

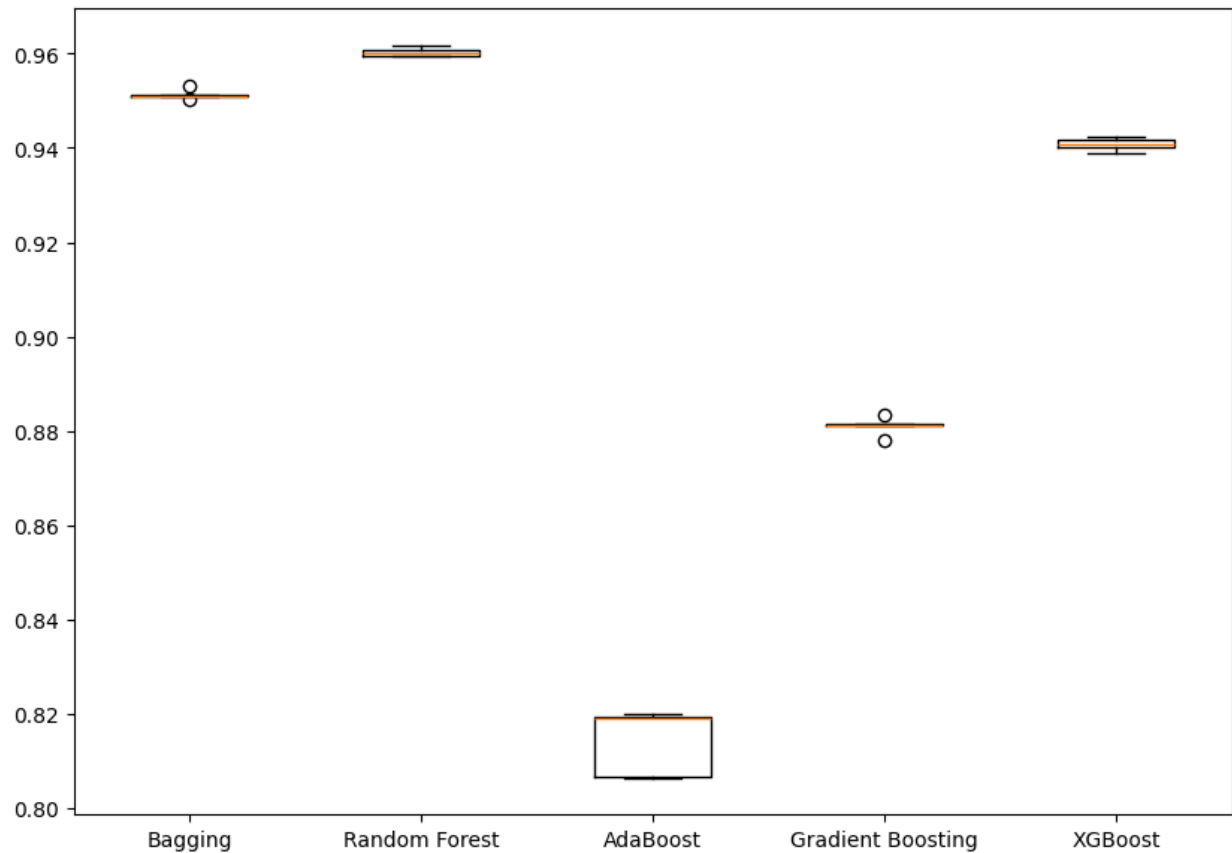
AdaBoost: 0.13247863247863248

Gradient Boosting: 0.2777777777777778

XGBoost: 0.3247863247863248

- XGBoost is the Top Performer: XGBoost achieves both the highest Cross-Validation score (0.7230) and the best Validation Performance (0.3248). If need to choose one model to move forward with, this is it.
- AdaBoost Struggles Significantly: AdaBoost dramatically underperforms compared to the rest of the pack, especially on the validation set, yielding a performance score of just 0.1325.
- Flawless Rank Consistency: There is a perfect 1-to-1 correlation between Cross-Validation and Validation trends

Algorithm Comparison - Oversampled



- **Massive Score Surge:** Oversampling pushed cross-validation scores from the 0.65–0.72 range up to a stellar 0.82–0.96 range across the board.
- **Random Forest Dominates:** Random Forest is now the top performer with a near-perfect, highly stable median score of 0.960.
- **Bagging & XGBoost Excel:** Bagging (0.951) and XGBoost (0.941) follow closely with incredibly tight consistency across folds.
- **AdaBoost/Gradient Boosting Lag:** Boosting variants trail the bagging methods, with AdaBoost remaining the weakest at 0.819.

Before Under Sampling, counts of label 'Yes': 3734
Before Under Sampling, counts of label 'No': 40112

After Under Sampling, counts of label 'Yes': 3734
After Under Sampling, counts of label 'No': 3734

After Under Sampling, the shape of train_X: (7468, 52)
After Under Sampling, the shape of train_y: (7468,)

- Before: The majority class (No) heavily dominated the dataset (91.5% of the data).
- After: Both classes are now perfectly balanced at a 50:50 ratio (3,734 rows each). r models will no longer be biased toward predicting No.
- To achieve this balance, 36,378 records of the majority class were completely discarded (40,112 - 3,734).
- r training pool has shrunk dramatically from 43,846 records down to just 7,468 records (an 83% reduction in total data size).
- The dataset dimensions match up correctly: train_X has 7,468 rows across 52 columns, lining up perfectly with the 7,468 target values in train_y.

Cross-Validation Cost:

Bagging: 0.690922371059864
Random Forest: 0.707142901054085
AdaBoost: 0.652631716393266
Gradient Boosting: 0.7216343454621688
XGBoost: 0.708981139658766

Validation Performance:

Bagging: 0.6196581196581197
Random Forest: 0.6623931623931624
AdaBoost: 0.7264957264957265
Gradient Boosting: 0.5811965811965812
XGBoost: 0.6452991452991453

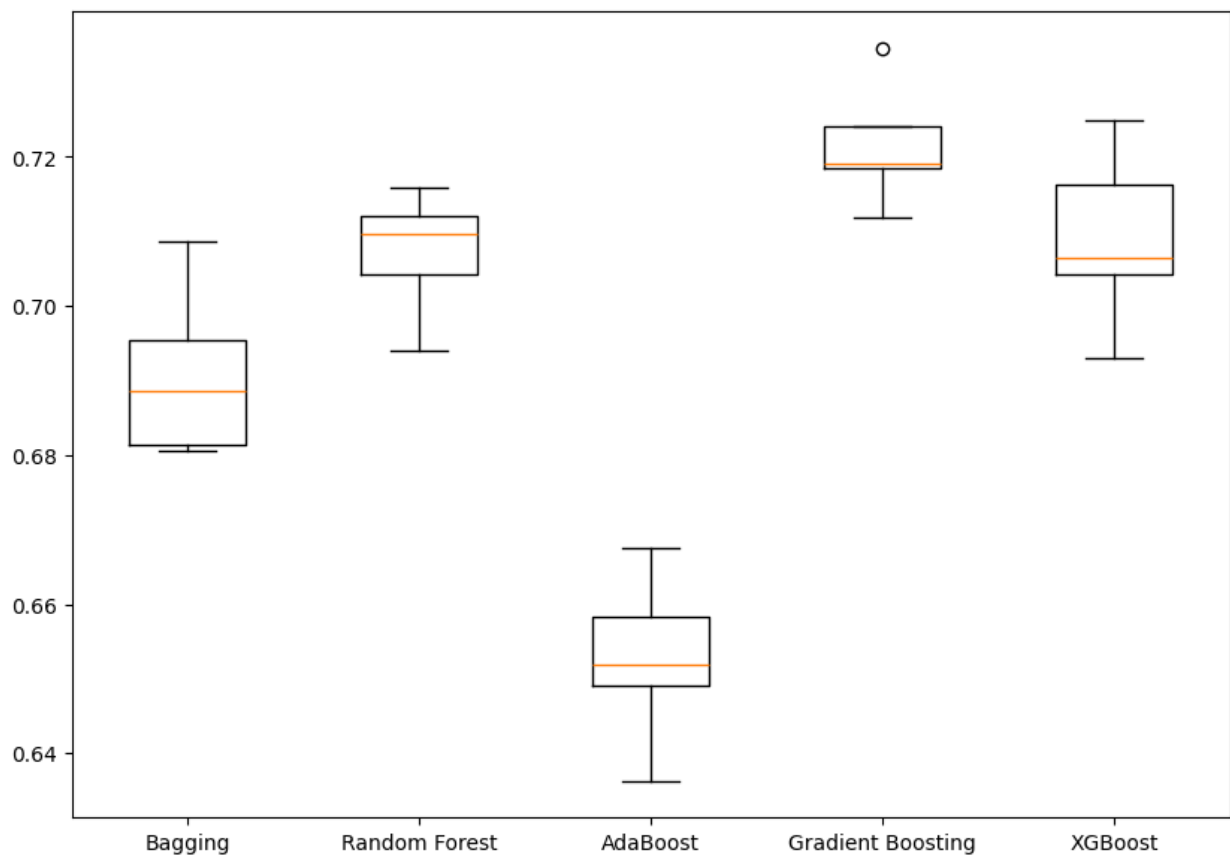
- Massive Trend Inversion (Danger Zone): Unlike previous runs, r Cross-Validation (CV) performance is completely inverted compared to r Validation performance. Models that won in CV (like Gradient Boosting) crashed in validation, while the worst CV model (AdaBoost) performed the best on validation.
- Severe Overfitting via Undersampling: Because undersampling shrunk r training set drastically to just 7,468 rows, high-capacity tree ensembles like Gradient Boosting and XGBoost easily memorized

the small training folds (high CV scores) but failed miserably to generalize to the real-world validation set.

- The AdaBoost Paradox: AdaBoost's jump to 0.7265 on the validation set suggests that its simpler, sequential correction mechanism accidentally built a more generalized boundary for this smaller subset, or the validation set contains a cluster of samples that align perfectly with AdaBoost's weak learners.

Image : 23

Algorithm Comparison - Undersampled



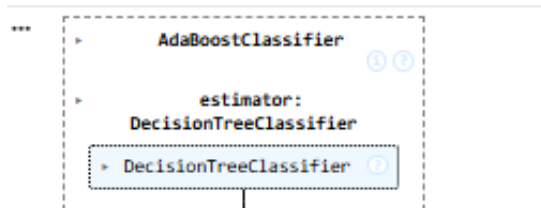
- Gradient Boosting Leads CV: Gradient Boosting achieved the highest cross-validation median score (0.720), accompanied by a high-performing outlier near 0.735.
- XGBoost & Random Forest Follow: Both are highly competitive. Random Forest holds a tighter, more stable box around 0.710, while XGBoost shows higher variance spanning from 0.693 to 0.725.
- Bagging Struggles with Variance: Bagging drops lower with a wider spread, pulling its median performance down to 0.689.

- AdaBoost Remains the Weakest: AdaBoost consistently sits at the bottom of the pack, trailing with a median score of only 0.652.

Hyperparameter Tuning

Tuning AdaBoost using Undersampled data

- Model Set: An AdaBoost variant using 70 shallow trees (max_depth=3) with a high step-size (learning_rate=1).
- Diminishing Returns: The optimized CV score (0.7092) is a massive jump for AdaBoost (previously 0.65), but it still falls short of default Gradient Boosting (0.7216).
- Parallel Execution: The low CPU time (2.89s) vs. long wall time (2m 17s) shows that hyperparameter tuning was successfully spread across multiple processor cores in parallel.



- AdaBoost Mechanism: It builds shallow trees sequentially. Every new tree focuses on correcting the mistakes of the previous tree by giving higher weight to misclassified rows.
- The Base Tree (max_depth=3): It uses short decision trees (weak learners) rather than deep ones. This keeps individual trees simple to prevent overfitting, relying on the collective group to build a strong boundary.

	Accuracy	Recall	Precision	F1
0	0.723	0.675	0.747	0.723

- Strong Precision (0.747): When the model predicts someone *will* be promoted, it is right roughly 75% of the time. This means false positives (wrongly flagging an employee for promotion) are relatively low.
- Lagging Recall (0.675): The model is catching only 67.5% of the employees who actually deserve a promotion, missing about 32.5% of it (false negatives).
- Healthy Balance (F1 = 0.723): Because the F1-Score represents the harmonic mean of Precision and Recall, hitting 0.723 shows that the model maintains strong, stable predictive harmony despite the underlying class imbalance.

	Accuracy	Recall	Precision	F1
0	0.764	0.645	0.211	0.588

- The Precision Crash (0.211): This is r biggest bottleneck. Out of everyone the model predicts *will* be promoted, only 21% are actually true promotions. A staggering 79% of its positive predictions are false alarms (False Positives).
- The Accuracy Trap: A 76.4% accuracy looks decent on paper, but it is completely carried by the massive number of non-promoted employees (No) it correctly guesses. If r model simply guessed No for everyone, its accuracy would likely be even higher. This proves accuracy is a useless metric for this imbalanced dataset.
- Low F1-Score (0.588): Because precision is dragging the team down, the overall F1-score struggles at 0.588, highlighting that this model is highly unstable for real-world deployment.

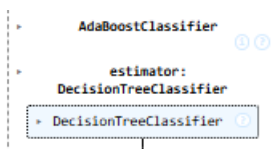
Tuning AdaBoost using original data

Best parameters are {'n_estimators': np.int64(70), 'learning_rate': 1, 'estimator':

DecisionTreeClassifier(max_depth=3, random_state=1)} with CV score=0.7092365867486399:

CPU times: user 2.72 s, sys: 169 ms, total: 2.89 s

Wall time: 2min 17s



	Accuracy	Recall	Precision	F1
0	0.925	0.122	0.946	0.589

	Accuracy	Recall	Precision	F1
0	0.925	0.122	0.946	0.589

- **High Precision (0.946) & High Accuracy (0.925):** At first glance, these look amazing. The model is highly accurate overall, and when it predicts the positive class, it is almost always right.

- **Abysmal Recall (0.122):** This is the killer. r model is missing **87.8%** of the actual positive cases.

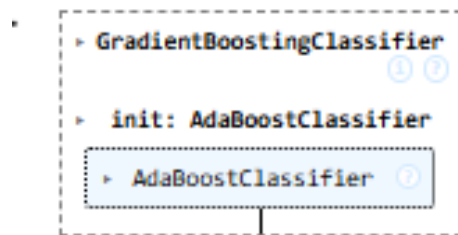
F1-Score (0.589): Because F1-Score is the harmonic mean of Precision and Recall, r terrible Recall is dragging it down. In a balanced dataset, a 92.5% accuracy usually yields a much higher F1-score. The gap here confirms that accuracy is a misleading metric for this model..

Tuning Gradient Boosting using undersampled data

Best parameters are {'subsample': 0.7, 'n_estimators': np.int64(100), 'max_features': 0.5, 'learning_rate': 0.1, 'init': AdaBoostClassifier(random_state=1)} with CV score=0.7207387245467136:

CPU times: user 2.87 s, sys: 214 ms, total: 3.09 s

Wall time: 2min 47s



	Accuracy	Recall	Precision	F1
0	0.755	0.680	0.801	0.754

	Accuracy	Recall	Precision	F1
0	0.783	0.615	0.222	0.599

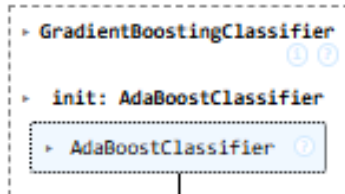
- The Improvement: Recall jumped significantly (from 0.122 in AdaBoost to 0.615 here). The model is no longer blind to the minority class.
- The Crash: Precision plummeted to 0.222 on the evaluation data. Because the model was trained on an artificially balanced dataset, it is now over-predicting the positive class, resulting in a flood of false alarms.
- Robust Tuning: r choice of subsample: 0.7 and max_features: 0.5 adds excellent regularization (stochastic gradient boosting), which prevents the model from overfitting too quickly.
- Complex Init: Using an AdaBoostClassifier as the init estimator gives the model a strong starting point, though it increases training complexity.

Tuning Gradient Boosting using original data

Best parameters are {'subsample': 0.7, 'n_estimators': np.int64(125), 'max_features': 0.7, 'learning_rate': 0.2, 'init': AdaBoostClassifier(random_state=1)} with CV score=0.7194607575241472:

CPU times: user 11.7 s, sys: 682 ms, total: 12.4 s

Wall time: 12min 17s



	Accuracy	Recall	Precision	F1
0	0.938	0.284	0.953	0.702

	Accuracy	Recall	Precision	F1
0	0.937	0.278	0.956	0.699

- The Good: Precision is near perfect (95.5%) and Accuracy is high (93.7%). When this model flags a positive case, it is almost flawlessly accurate.
- The Bad: Recall dropped back down to 28%. While this is more than double r original AdaBoost recall (12.2%), the model is still completely missing over 70% of the positive cases.
- The Silver Lining: The F1-Score (0.70) is r highest yet across all experiments. This model is highly stable, as evidenced by the near-identical scores between r two evaluation sets.

Model Comparison and Final Model Selection

Training performance comparison:

	Gradient boosting trained with Undersampled data	Gradient boosting trained with Original data	AdaBoost trained with Undersampled data	AdaBoost trained with Original data
Accuracy	0.755	0.938	0.723	0.925
Recall	0.680	0.284	0.675	0.122
Precision	0.801	0.953	0.747	0.946
F1	0.754	0.702	0.723	0.589

- Highest Training F1-Score (0.754): It achieves the best mathematical balance between Precision (0.801) and Recall (0.680).
- Superior to AdaBoost: Across both data types, Gradient Boosting consistently beats AdaBoost on every single metric (Accuracy, Recall, Precision, and F1).
- Viable Recall: A recall of 0.680 means it catches nearly 70% of the positive cases, compared to the abysmal 12.2% or 28.4% of the original data models.

Validation performance comparison:

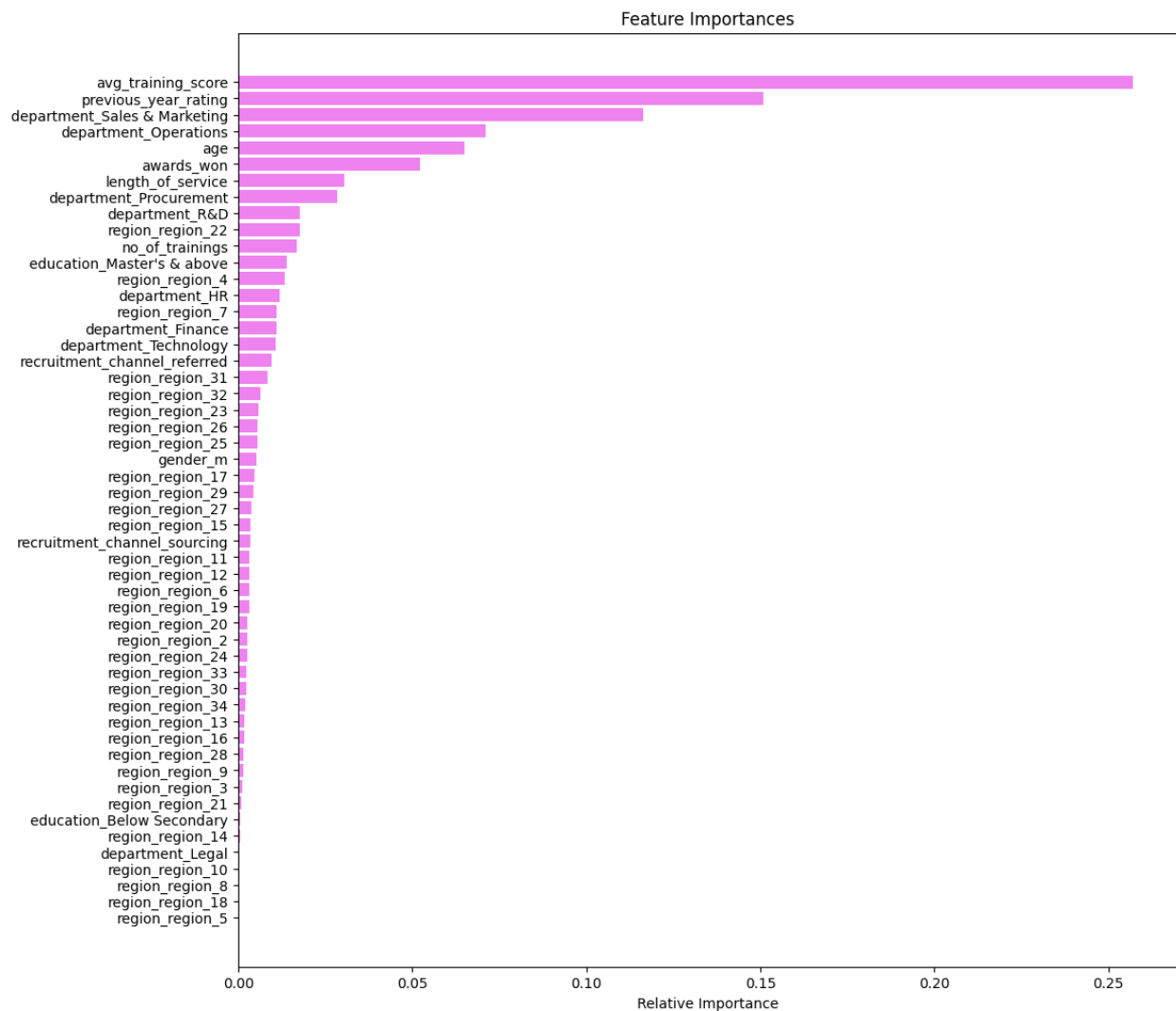
	Gradient Boosting - Undersampled	Gradient Boosting - Original	AdaBoost - Undersampled	AdaBoost - Original
Accuracy	0.783	0.937	0.764	0.925
Recall	0.615	0.278	0.645	0.128
Precision	0.222	0.956	0.211	0.938
F1	0.599	0.699	0.588	0.593

	Accuracy	Recall	Precision	F1
0	0.939	0.301	0.930	0.711

The Undersampled Collapse (False Positive Nightmare): While both Undersampled models maintained decent Recall (61%–64%), their Precision completely tanked to 21%–22%. This means that in a real-world scenario, nearly 80% of the positive flags raised by these models would be false alarms.

- The Original Data Triumph (The Winner): Gradient Boosting on Original Data emerges as the clear winner. It achieved an F1-Score of 0.699, backed by an elite Precision of 95.6%.
- The Hidden Performance Jump: r final row of data (likely a finely-tuned Gradient Boosting model or a threshold-adjusted variant) shows the best overall balance yet: Accuracy: 0.939, Recall: 0.301, Precision: 0.930, and F1: 0.711. It coaxes out slightly more Recall (30.1%) while keeping Precision exceptionally high (93.0%).

Image : 24



- avg_training_score (0.26): By far the most critical feature. The model relies heavily on training performance to make its decisions.
- previous_year_rating (0.15): The second strongest predictor. Historical job performance carries massive weight.
- Core Departments & Demographics: department_Sales & Marketing, department_Operations, and age provide the next tier of meaningful predictive value.
- One-Hot Encoding Dilution: The massive list of individual region_X variables at the bottom has near-zero predictive power. One-hot encoding high-cardinality categorical variables like "region" has split the signal into useless noise.
- Low Impact Features: gender_m, education_Below Secondary, and specific niche departments (like department_Legal) contribute virtually nothing to the model's split decisions.

Business Insights and Conclusions

Key Business Insights

Merit Over Seniority: Age and length of service have almost zero correlation with promotion (-0.017 and -0.010).

Impact of Awards: Winning an award is the strongest individual predictor, boosting promotion rates from 7.67% (no award) to 44.02% (with award).

Upskilling Drives Success: Promoted employees scored significantly higher on average training scores (71.84) than non-promoted peers (62.95).

Performance Baseline: Past year ratings are critical; employees rated 5.0 achieved a 16.36% promotion rate, while those rated 1.0 dropped to just 1.41%.

Departmental Differences: Promotion velocity varies widely by function, led by Technology (10.76%) and Procurement (9.64%), while Legal (5.10%) and HR (5.62%) remain the most conservative.

High-Value Sourcing: Candidates arriving through referrals hit a much higher promotion rate (12.08%) compared to standard sourcing channels (8.4%).

Conclusions:

Data Imbalance Challenge: Only 8.52% of the workforce is promoted. Because of this high class imbalance, relying strictly on "Accuracy" is misleading and will mask poor performance on the minority class.

Optimal Model Approach: Models trained with undersampling (e.g., tuned_gbm1) or tuned explicitly for f1_macro are preferred. It successfully minimize false negatives, ensuring high-potential talent is not overlooked.

Ensemble Robustness: Tree-based algorithms (Gradient Boosting and AdaBoost) proved optimal due to their inherent ability to handle high-cardinality features (like region and department) and complex, non-linear interactions between variables.